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Learning to Align

Addressing Frequency Distribution Shifts

John Pavlopoulos
Assistant Professor, Athens University of Economics and Business
Archimedes, Athena Research Center, Greece

Panagiotis Kaliosis
PhD student at Stony Brook University

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- What is **F**requency **A**ware **D**istribution **A**lignment
- Theoretical aspects of FADA
- Application to the Humanities
- Application to Healthcare (if time permits)

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FADA: Why → *models learn dataset biases*

- Models learn the wrong priors

FADA: Why → *real domains have different statistics*

- Models learn the wrong priors
- Distribution shifts in real data
  - historical documents
  - different genres / authors
  - different languages sharing the same alphabet

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FADA: Why → *real domains have different statistics*

- Models learn the wrong priors
- Distribution shifts in real data
  - historical documents
  - different genres / authors
  - different languages sharing the same alphabet

Token	Training	Target
A	50%	20%
B	30%	40%
C	20%	40%

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FADA: Why → *Use Global Distribution Constraints*

- Models learn the wrong priors
- Distribution shifts in real data
- Predictions Should Be Globally Plausible
  - Standard decoding maximizes likelihood
  - Ignoring global statistics of the target domain

Predictions should not only be "locally likely,"
but also globally consistent with the statistics of the target domain.

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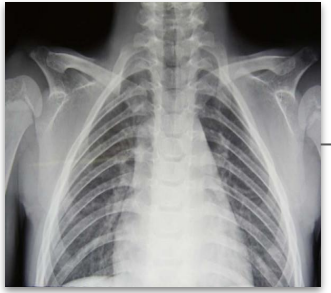
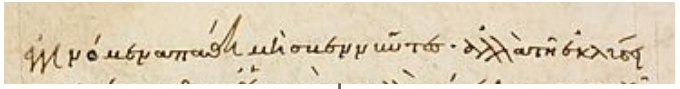
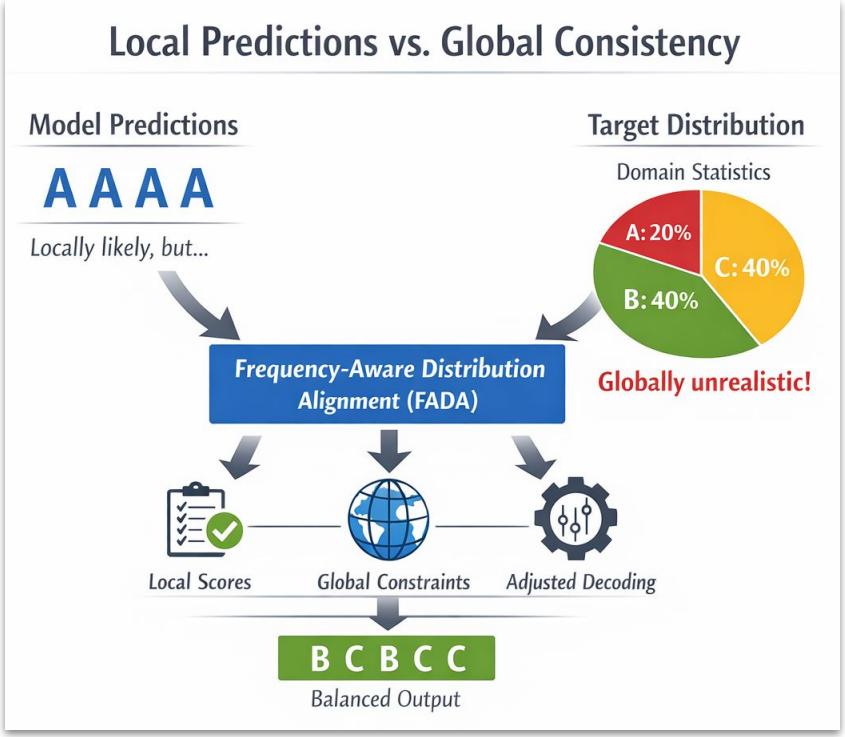
FADA: Why



Chest X-ray showing bilateral clavicular hypoplasia…



ἐγγινομένα πάθη μὴ σβεννύντες ἀλλὰ τῇ εκλύσει



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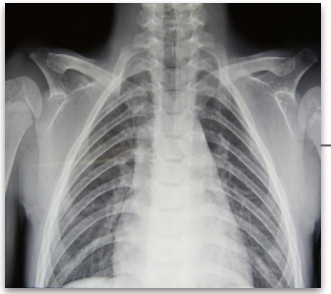
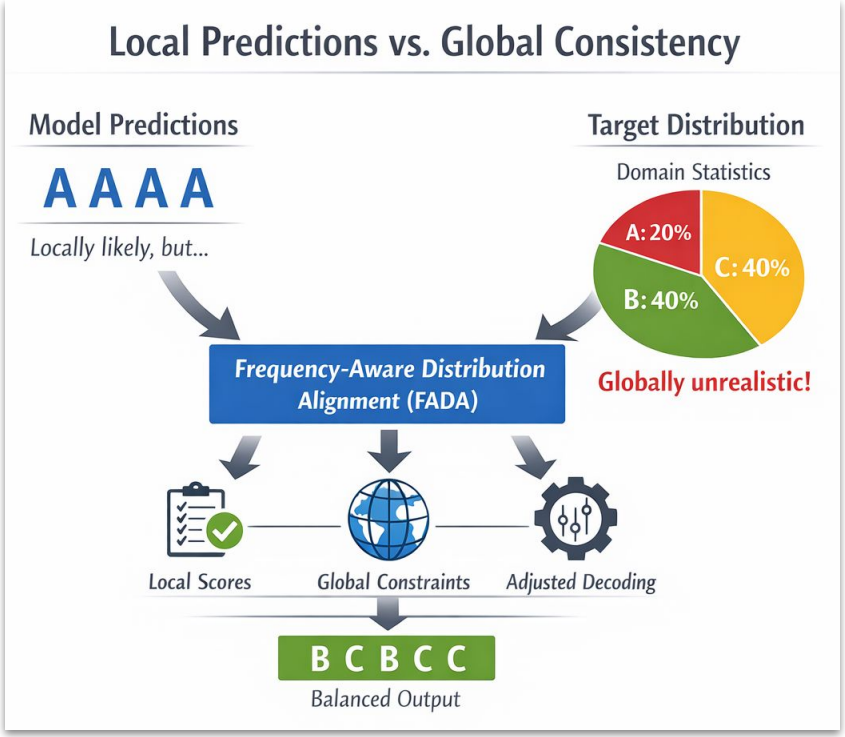


Chest X-ray showing bilateral clavicular hypoplasia…

normal,
normal,
mal,
mal, …



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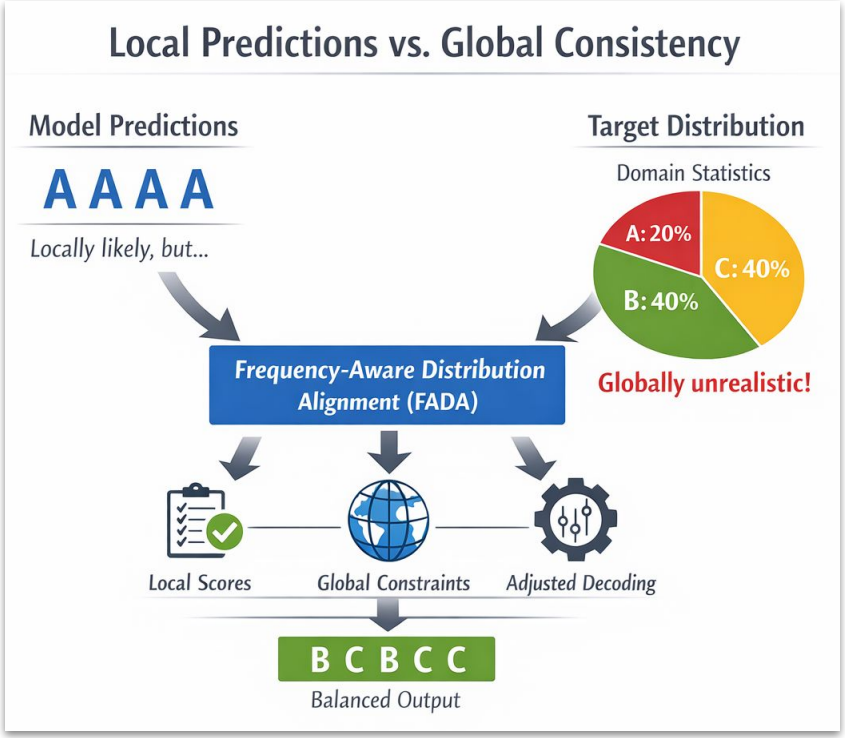

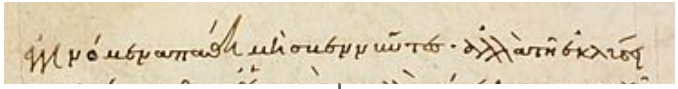

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ااااااا

ἐγγγγγγγγγγγγγγ

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How information about the test instance can improve model accuracy

The Research Question

Can we handle intra-dataset distributional variation to let a model ***learn general task-relevant representations*** *while adjusting its output to* ***reflect expected patterns of the specific input context*** *without requiring separate adaptation stages or domain boundaries*

- *handle intra-dataset distributional variation*
- *learn general task-relevant representations*
- *reflect expected patterns of the input*
- *no parameter updating*

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What is FADA

- x : input (image, text, signal, etc.)
- $y = (y_1, \ldots, y_T)$: predicted sequence
- $y_t \in V$ where V is a vocabulary

$$P_\theta(y|x) = \prod_{t=1}^{T} P_\theta(y_t \mid y_{<t}, x)$$

$$\mathcal{L}_{NLL} = -\sum_{t=1}^{T} \log P_\theta(y_t^* \mid y_{<t}^*, x)$$

where y^* is the ground truth.

What is FADA

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$$\hat{p}_y(v) = \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}(y_t = v)$$

A B B C $\longrightarrow$ $\hat{p}_y = (A : 0.25,\ B : 0.5,\ C : 0.25)$

an empirical distribution

(Target distribution prior)

How information about the test instance can improve model accuracy

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$$p_{target}(v)$$

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$$p_{target}(v)$$

$$D(\hat{p}_y, p_{target})$$

encourage predictions whose ***token frequencies match this distribution***

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How information about the test instance can improve model accuracy

What is FADA

- x : input (image, text, signal, etc.)
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High $D \Rightarrow$ high loss

$$\boxed{\mathcal{L}} = \mathcal{L}_{NLL} + \lambda \boxed{D(\hat{p}_{y^{pred}}, p_{target})}$$

$$\mathcal{L}_{NLL} = -\sum_{t=1}^{T} \log P_\theta(y_t^* \mid y_{<t}^*, x)$$

$$p_{target}(v) \longrightarrow (A : 0.25,\ B : 0.5,\ C : 0.25)$$

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What is FADA

- x : input (image, text, signal, etc.)
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$$\mathcal{L} = \mathcal{L}_{NLL} + \lambda D(\hat{p}_{y^{pred}}, p_{target})$$

$$D(\hat{p}_\theta, p_{target})$$

$$\hat{p}_y(v) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}(y_t = v)$$

$$\hat{p}_\theta(v) = \frac{1}{T}\sum_{t=1}^{T} P_\theta(y_t = v | y_{<t}, x)$$

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What is FADA

- x : input (image, text, signal, etc.)
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- $y_t \in V$ where V is a vocabulary

$$y^* = \arg\max_y \log P_\theta(y|x)$$

search for the token with the highest probability.

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beam search prefers sequences that are both, likely under the model and match the expected distribution.

$$score(y) = \log P_\theta(y|x) - \lambda D(\hat{p}_y, p_{target})$$

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What is FADA

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$$y^* = \arg\min_y E(y, x)$$

$$E(y, x) = -\log P_\theta(y|x) + \lambda D(\hat{p}_y, p_{target})$$

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(What the method is really doing)

How information about the test instance can improve model accuracy

What is FADA

- x : input (image, text, signal, etc.)
- $y = (y_1, \ldots, y_T)$: predicted sequence
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$$\hat{p}_y \approx p_{target}$$

...maximize likelihood subject to distribution alignment

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(General interpretation)

How information about the test instance can improve model accuracy

What is FADA

- x : input (image, text, signal, etc.)
- $y = (y_1, \ldots, y_T)$: predicted sequence
- $y_t \in V$ where V is a vocabulary

expected statistic

$$\arg\max_{y} \left(\log P_\theta(y|x) - \lambda D(f(y), c)\right)$$

global statistics extracted from the sequence

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(General interpretation)

How information about the test instance can improve model accuracy

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global statistics extracted from the sequence

In fada, f(y) is the token frequency vector

- *sentiment distribution*
- *topic proportions*
- *POS tag frequencies*
- *entity types*
- *event distributions.*

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(General interpretation: Langragian)

How information about the test instance can improve model accuracy

What is FADA

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$$\arg\max_y \left(\log P_\theta(y|x) - \lambda D(f(y), c)\right)$$

$$\max_y \log P_\theta(y|x) \quad s.t. \quad f(y) = c$$

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What is FADA

- x : input (image, text, signal, etc.)
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$$f_{v,w}(y) = \frac{1}{T} \sum_t \mathbf{1}(y_t = v, y_{t+1} = w)$$

- Matching only 1st order moments, ignoring higher-order dependencies

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What is FADA

- x : input (image, text, signal, etc.)
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- Matching only 1st order moments: ignoring higher-order dependencies
- Sensitivity to sequence length: the constraint becomes noise

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- Matching only 1st order moments: ignoring higher-order dependencies
- Sensitivity to sequence length: the constraint becomes noise
- Incorrect priors can hurt performance: small samples, noise, partial data

What is FADA

- x : input (image, text, signal, etc.)
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- $y_t \in V$ where V is a vocabulary

bigram distribution

$$score(y) = \log P_\theta(y|x) - \lambda_1 D(f_1(y), c_1) - \lambda_2 D(f_2(y), c_2)$$

How information about the test instance can improve model accuracy

What is FADA

$$\mathcal{S}_{\text{GD}} = [\lambda \cdot \sum_{i=1}^{t} \log P(v_i | v_{<i}) - (1-\lambda) \cdot \boxed{W_2(D_{\text{pred}}, D_{\text{emp}})}]$$

$$\mathcal{L}_{W_2} = W_2(D, Q) = \left(\frac{1}{n} \sum_{i=1}^{n} \| D_{(i)} - Q_{(i)} \|^2 \right)^{\frac{1}{2}}$$

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- Alignment without FAD
- Discussion and future directions

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Motivation

- Language **evolves** across *centuries* / between *regions*.

- New characters are introduced & others become extinct.
  - Why? Due to linguistic evolution, orthographic reforms or just contextual usage shifts.

- New dialects were created due to population movements / tone system changes

IF: ἕ and ῶ appear frequently in 10CE, but not in 12CE
THEN: How would HTR trained on mixed data cope with this?

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Suboptimal

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Suboptimal

Not an explored problem

Motivation

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- New characters are introduced & others become extinct.
  - Why? Due to linguistic evolution, orthographic reforms or just contextual usage shifts.

- New dialects were created due to population movements / tone system changes

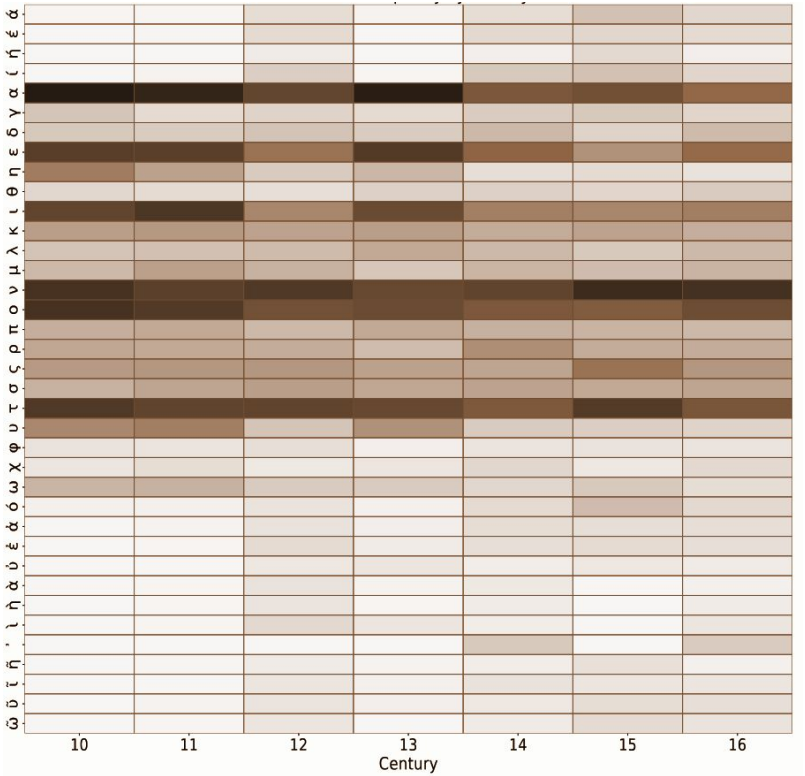
- Character frequency distributions **change across time / regions**

- Training new models for “versions” of the same language is **not optimal**:
  not enough data / need to retrain from scratch for new datasets.

Data Exploration

- Recognition models inherently learn to model the training data distributions
- BUT… what happens when different subsets of the data exhibit different distributions?

color intensity ~ high frequency

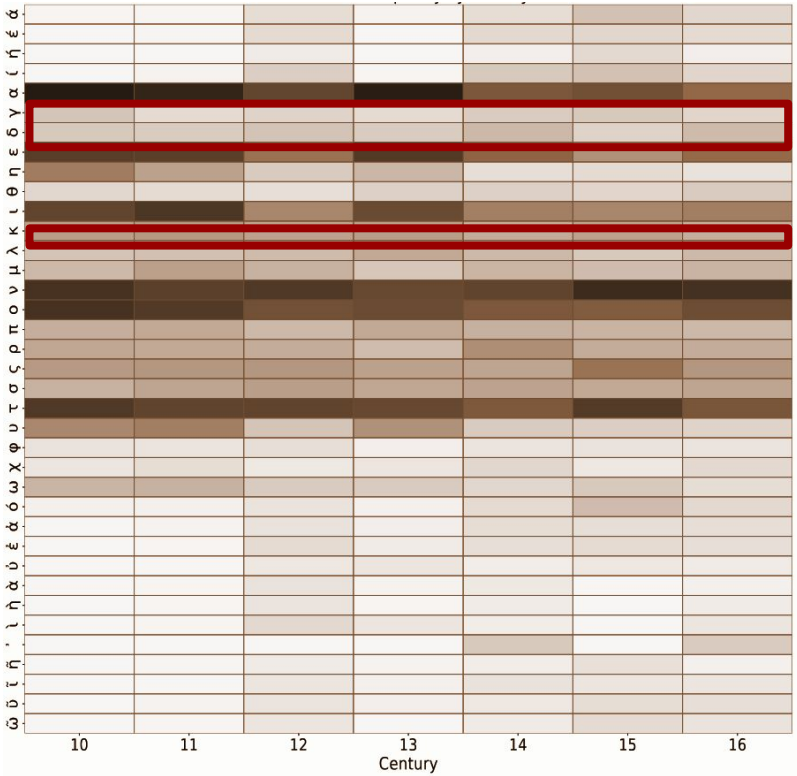


Paraskevi Platanou, John Pavlopoulos, and Georgios Papaioannou. 2022. Handwritten paleographic Greek text recognition: A century-based approach. In LREC, pages 6585–6589, Marseille, France. ELRA.

Data Exploration

Stable across centuries

color intensity ~ high frequency



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Data Exploration

Varying across centuries

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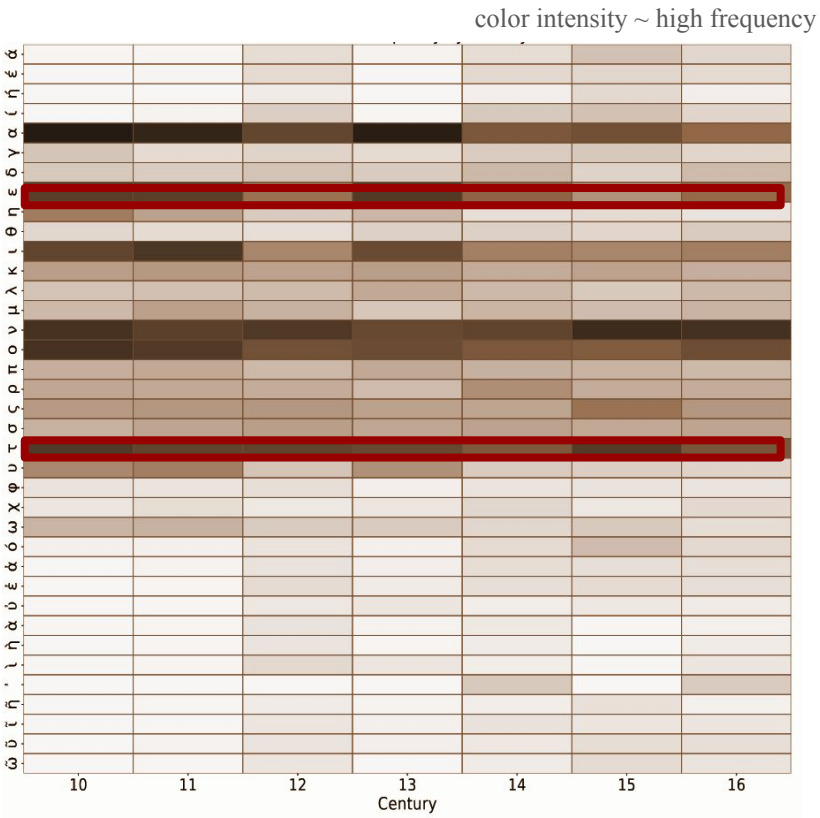

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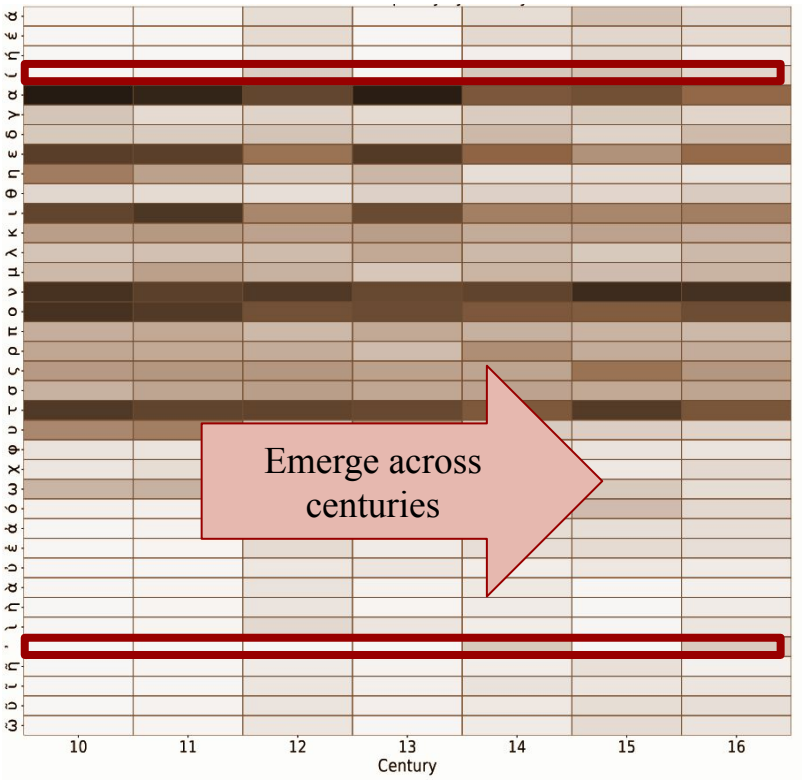

FADA: The proposed method

The core idea:

align *predicted* vs. *empirical* character frequency distributions **(CFDs)**

The goal:

Reduce discrepancies between the expected and the predicted **CFDs**
usually caused by temporal and contextual shifts

"If a model predicts that 'ε' appears 5% of the time for a given century, but the character is actually used 12% in that century, we add a small penalty — a nudge — to correct that imbalance."

FADA serves as a **character-level** distribution alignment framework:

1. Training-time alignment → actively changing the model weights to adapt to empirical priors.
2. Inference-time alignment → guide the model towards distributionally aligned transcriptions during inference.

FADATR: Training-time Alignment

How:

1. Calculate the priors: Compute the empirical character frequency distributions for each century.

2. At each training step, introduce an auxiliary distribution alignment loss:

   - Compute the model's predicted character frequency distributions.

   - Calculate the second-order Wasserstein distance between the empirical (expected) and the predicted century-level character frequency distributions.

$$\mathcal{L}_{W_2} = W_2(D, Q) = \left(\frac{1}{n} \sum_{i=1}^{n} \| D_{(i)} - Q_{(i)} \|^2 \right)^{\frac{1}{2}}$$

Note!! This is a *complementary* loss to the primary recognition objective (CTC or CE) → designed to enhancing the model's robustness to intra-dataset distributional shifts

FADAGD: Inference-time Alignment

What: An additional mechanism for refining predictions **without** requiring model retraining (**G**uided **D**ecoding)

How: Incorporate a frequency-aware scoring component into the beam search process, which biases candidate sequences toward those whose character distributions more closely match the expected empirical ones.

At each decoding step: each candidate sequence V = (v1, v2, ..., vt) is scored using the cumulative log-probs **AND** the second-order Wasserstein distance distribution alignment objective:

$$\mathcal{S}_{\mathrm{GD}} = [\lambda \cdot \sum_{i=1}^{t} \log P(v_i | v_{<i}) - (1-\lambda) \cdot W_2(D_{\mathrm{pred}}, D_{\mathrm{emp}})]$$

Desired outcome: Candidate sequences with both high likelihood and strong agreement with the expected character frequency distribution are favored during decoding

FADA: Experiments & Results

Handwritten Text Recognition (HTR) → main task of experimentation

- Two HTR models (TrOCR, C-RNN)
- Three datasets (one synthetic), Greek and French, 10-16 CE

Dataset	Model	Standard FT		FT w/ TR-TT		FT w/ $FADA_{TR\text{-}GD}$ (Ours)	
		CER ↓	WER ↓	CER ↓	WER ↓	CER ↓	WER ↓
HPGT	TrOCR	26.92 (0.96)	73.82 (1.31)	26.89 (0.61)	72.53 (0.81)	**25.06** (0.91)	70.14 (0.48)
	C-RNN	24.72 (0.38)	77.08 (0.85)	24.41 (0.55)	76.66 (0.81)	**23.89** (0.34)	76.31 (0.77)
CATMuS	TrOCR	**9.56** (0.22)	37.02 (0.28)	9.68 (0.17)	37.01 (0.69)	9.63 (0.25)	37.05 (0.47)
	C-RNN	17.08 (0.29)	54.79 (0.32)	27.12 (1.13)	67.61 (1.98)	**16.79** (0.52)	53.65 (1.02)
Synthetic	TrOCR	5.94 (0.09)	17.61 (0.31)	5.48 (0.17)	17.12 (1.96)	**5.19** (0.27)	16.60 (0.45)
	C-RNN	9.71 (0.73)	31.58 (2.73)	9.82 (0.34)	34.58 (1.78)	**8.31** (0.12)	27.08 (0.36)

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FADA: Experiments & Results

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  - Standard Fine-tuning (FT), following the popular path
  - FT + [special] token learning (FT w/TR-TT)

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FADA: Experiments & Results

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FADA$_{\text{TR-GD}}$ (both training and inference-time alignment) outperforms both baselines in most cases!

Dataset	Model	Standard FT		FT w/ TR-TT		FT w/ FADA$_{\text{TR-GD}}$ (Ours)	
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FADA: Experiments & Results

Automatic Speech Recognition (ASR) ⟶ secondary task of experimentation

- One ASR model (Whisper)
- One dataset (EdAcc, English) ⟶ speech samples from different English dialects

US English
Southern British English
Irish English
Indian English

- Same baseline setups
  - Standard Fine-tuning (FT), following the popular path
  - FT + [special] token learning (FT w/TR-TT)

FADA$_{\text{TR-GD}}$ (both training and inference-time alignment) outperforms both baselines!

Lower ↓ is better

Model	Standard FT	w/ TR-TT	w/ FADA$_{\text{TR-GD}}$
Whisper	27.11 / 40.02	28.88 / 44.09	**26.70** / 38.02

FADA: Ablation Study

- Should we change the model weights? → training-time alignment **(FADA$_{TR}$)**
- Should we simply rely on guiding the model towards better decoding paths at test time → inference-time alignment (FADA$_{GD}$)
- Should we do both? → training- & inference-time alignment **(FADA$_{TR\text{-}GD}$)**

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- Should we simply rely on guiding the model towards better decoding paths at test time → inference-time alignment (FADA$_{GD}$)
- Should we do both? → training- & inference-time alignment **(FADA$_{TR\text{-}GD}$)**

Dataset	Model	FADA$_{GD}$
HPGT	TrOCR	−1.4%
	C-RNN	−0.8%
CATMuS	TrOCR	+1.7%
	C-RNN	−0.4%
Synthetic	TrOCR	−0.2%
	C-RNN	−0.1%

Lower ↓ is better

FADA: Ablation Study

- Should we change the model weights? → training-time alignment **(FADA$_{TR}$)**
- Should we simply rely on guiding the model towards better decoding paths at test time → inference-time alignment (FADA$_{GD}$)
- Should we do both? → training- & inference-time alignment **(FADA$_{TR\text{-}GD}$)**

Dataset	Model	FADA$_{GD}$	FADA$_{TR}$
HPGT	TrOCR	−1.4%	−6.5%
	C-RNN	−0.8%	−3.2%
CATMuS	TrOCR	+1.7%	+1.6%
	C-RNN	−0.4%	−1.1%
Synthetic	TrOCR	−0.2%	−10.6%
	C-RNN	−0.1%	−12.9%

Lower ↓ is better

FADA: Ablation Study

- Should we change the model weights? → training-time alignment **(FADA$_{TR}$)**
- Should we simply rely on guiding the model towards better decoding paths at test time → inference-time alignment (FADA$_{GD}$)
- Should we do both? → training- & inference-time alignment **(FADA$_{TR\text{-}GD}$)**

Dataset	Model	FADA$_{GD}$	FADA$_{TR}$	FADA$_{TR\text{-}GD}$
HPGT	TrOCR	−1.4%	−6.5%	**−6.9%**
	C-RNN	−0.8%	−3.2%	**−3.3%**
CATMuS	TrOCR	+1.7%	+1.6%	**+0.7%**
	C-RNN	−0.4%	−1.1%	**−1.7%**
Synthetic	TrOCR	−0.2%	−10.6%	**−12.6%**
	C-RNN	−0.1%	−12.9%	**−14.4%**

Lower ↓ is better

FADA: Ablation Study

- Should we change the model weights? → training-time alignment **(FADA$_{TR}$)**
- Should we simply rely on guiding the model towards better decoding paths at test time → inference-time alignment (FADA$_{GD}$)
- Should we do both? → training- & inference-time alignment **(FADA$_{TR\text{-}GD}$)**

Dataset	Model	FADA$_{GD}$	FADA$_{TR}$	FADA$_{TR\text{-}GD}$
HPGT	TrOCR	−1.4%	−6.5%	**−6.9%**
	C-RNN	−0.8%	−3.2%	**−3.3%**
CATMuS	TrOCR	+1.7%	+1.6%	**+0.7%**
	C-RNN	−0.4%	−1.1%	**−1.7%**
Synthetic	TrOCR	−0.2%	−10.6%	**−12.6%**
	C-RNN	−0.1%	−12.9%	**−14.4%**

Lower ↓ is better

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- Should we change the model weights? → training-time alignment **(FADA$_{TR}$)**
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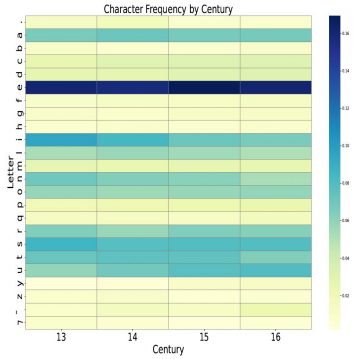

Dataset	Model	FADA$_{GD}$	FADA$_{TR}$	FADA$_{TR\text{-}GD}$
HPGT	TrOCR	−1.4%	−6.5%	**−6.9%**
	C-RNN	−0.8%	−3.2%	**−3.3%**
CATMuS	TrOCR	+1.7%	+1.6%	**+0.7%**
	C-RNN	−0.4%	−1.1%	**−1.7%**
Synthetic	TrOCR	−0.2%	−10.6%	**−12.6%**
	C-RNN	−0.1%	−12.9%	**−14.4%**

Lower ↓ is better

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FADA: Qualitative Results

Lower ↓ is better

Method	Transcription	CER
Baseline (TrOCR)	φωνης εἰσπνατιον τὴν δὲ κας ην συμφόνας ἀφηκαν τήν	29.16
FADA_{GD}	φωνης εἰσπνατιον τὴν δὲ κας ην συμφόνας ἀφηκαν τήν	29.16
FADA_{TR}	φωνης εἰσπνατττονται δίκας ἡν συμφώνως ἀφη καντὴν	14.58
$\text{FADA}_{\text{TR-GD}}$	φωνης εἰσπνατττονται δίκας ἡν συμφώνως ἀφη καντὴν	14.58
Ground Truth	φωνῆς εἰσπράττονται δίκας ἣν συμφώνως ἀφῆκαν τήν	
Baseline (Whisper)	yeah and all i bet and like in the apartment that we have	18.18
FADA_{GD}	and uh i bet in like in the apartment that we have	19.98
FADA_{TR}	and uh i bet in like in the apartment that we have	19.98
$\text{FADA}_{\text{TR-GD}}$	yeah i know i bet in like in the apartment that we have	**3.63**
Ground Truth	yeah i know i bet uh like in the apartment that we have	

FADA: Qualitative Results

Lower ↓ is better

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Baseline (TrOCR)	φωνης εἰσπνατιον τὴν δὲ κας ην συμφόνας ἀφηκαν τήν	29.16
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Thank you!

QA

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Appendix

- Discussion
- Data exploration
- Application to Healthcare

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Discussion over the *Findings*

1. **$\text{FADA}_{\text{GD}} < \text{FADA}_{\text{TR}}$**

Dataset	Model	FADA_{GD}	FADA_{TR}
HPGT	TrOCR	−1.4%	−6.5%
	C-RNN	−0.8%	−3.2%
CATMuS	TrOCR	+1.7%	+1.6%
	C-RNN	−0.4%	−1.1%
Synthetic	TrOCR	−0.2%	−10.6%
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Discussion over the *Findings*

1. **$FADA_{GD} < FADA_{TR}$**
   a. lightweight
   b. no re-training
   c. low-resource-friendly

Dataset	Model	$FADA_{GD}$	$FADA_{TR}$
HPGT	TrOCR	−1.4%	−6.5%
	C-RNN	−0.8%	−3.2%
CATMuS	TrOCR	+1.7%	+1.6%
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Discussion over the *Findings*

Lower ↓ is better

1. **$FADA_{GD}$ < $FADA_{TR}$**
   a. lightweight
   b. no re-training
   c. low-resource-friendly
   d. post-hoc gains

Dataset	Model	$FADA_{GD}$	$FADA_{TR}$	$FADA_{TR\text{-}GD}$
HPGT	TrOCR	−1.4%	−6.5%	**−6.9%**
	C-RNN	−0.8%	−3.2%	**−3.3%**
CATMuS	TrOCR	+1.7%	+1.6%	**+0.7%**
	C-RNN	−0.4%	−1.1%	**−1.7%**
Synthetic	TrOCR	−0.2%	−10.6%	**−12.6%**
	C-RNN	−0.1%	−12.9%	**−14.4%**

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1. **$\text{FADA}_{\text{GD}} < \text{FADA}_{\text{TR}}$**
2. Interpretability gains (compared to context conditioning)
3. vs. Domain Adaptation

source $\longrightarrow$ target

Discussion over the *Findings*

1. **$FADA_{GD} < FADA_{TR}$**
2. Interpretability gains (compared to context conditioning)
3. vs. Domain Adaptation
    a. not transferring from source to target

Discussion over the *Findings*

1. **$FADA_{GD} < FADA_{TR}$**
2. Interpretability gains (compared to context conditioning)
3. vs. Domain Adaptation
   a. not transferring from source to target
   b. task-relevant representations, adjusted to patterns

Discussion over the *Findings*

1. **$FADA_{GD} < FADA_{TR}$**
2. Interpretability gains (compared to context conditioning)
3. vs. Domain Adaptation
4. Computational overhead

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1. **$FADA_{GD}$ < $FADA_{TR}$**
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4. Computational overhead
   a. **$FADA_{TR}$** +5% in training time (**+0% at inference**)

Discussion over the *Findings*

1. **$FADA_{GD} < FADA_{TR}$**
2. Interpretability gains (compared to context conditioning)
3. vs. Domain Adaptation
4. Computational overhead
   a. $FADA_{TR}$ +5% in training time (+0% at inference)
   b. $FADA_{GD}$ +8% at inference (**+0% in training**)

Closing remarks

1. FADA improves robustness to distribution shifts

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2. Effective across modalities, languages, and architectures

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1. FADA improves robustness to distribution shifts
2. Effective across modalities, languages, and architectures
3. Future: extend to morphosyntactic or semantic distributions

Appendix

- Discussion
- **Data exploration**
- Application to Healthcare

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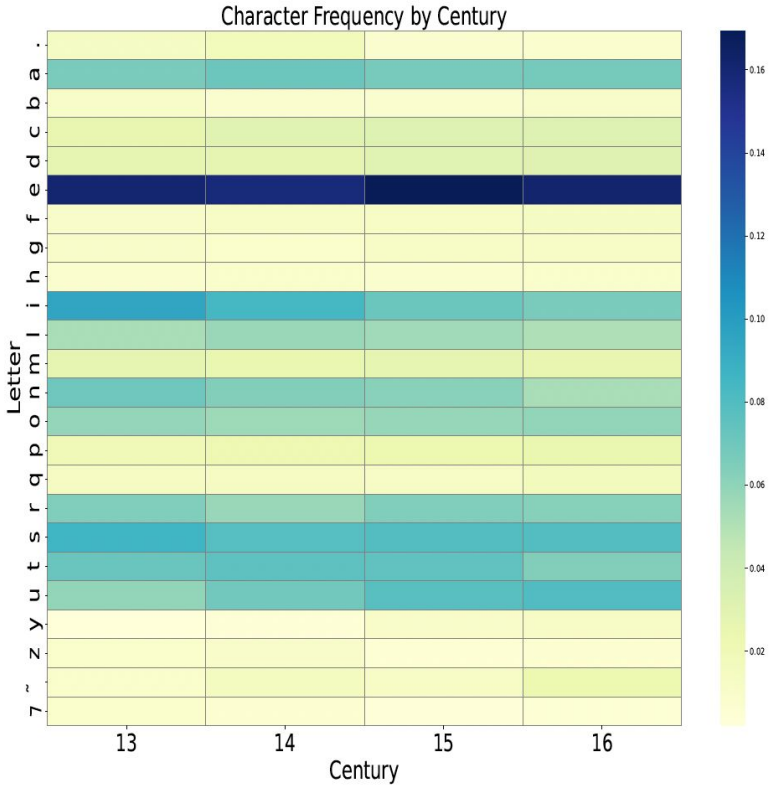
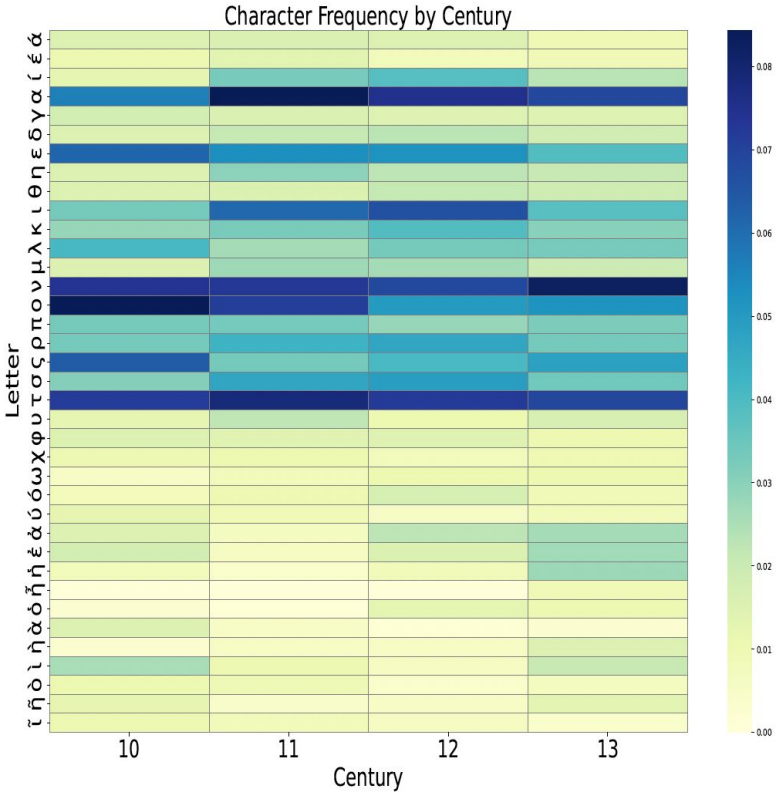
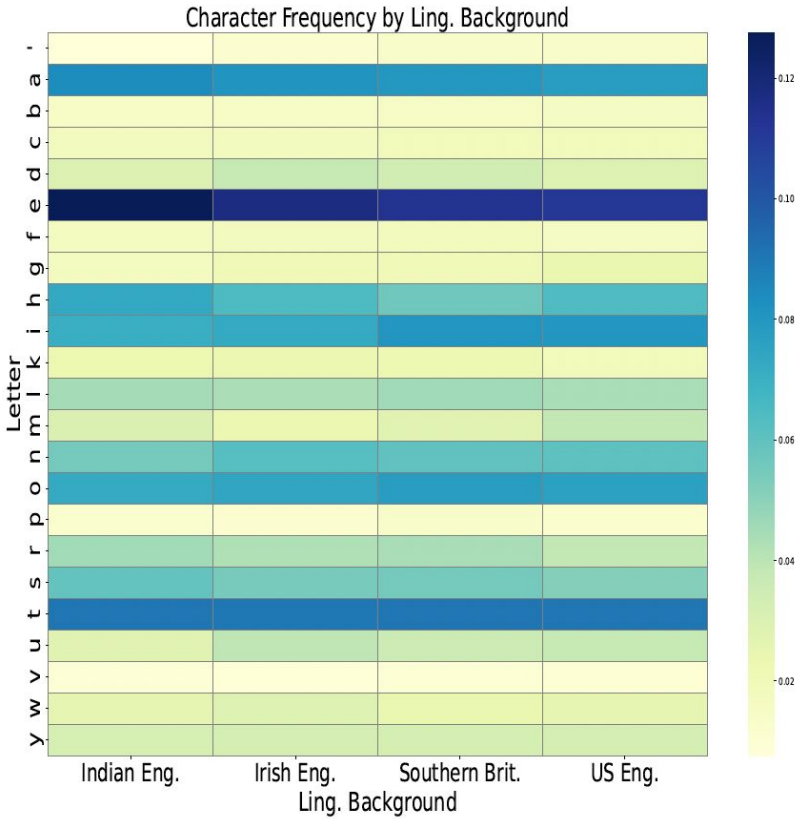
Exploration: CATMuS



Exploration: Synthetic data



Exploration: EdAcc



Appendix

- Discussion
- Data exploration
- **Application to Healthcare**

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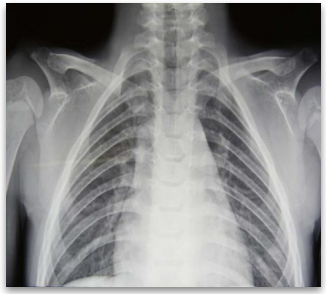
FADA in Diagnostic Captioning

Captioning: given a radiology image, generate *a draft diagnostic report*.



Generate caption…

Chest X-ray showing bilateral clavicular hypoplasia…

Tagging: given a radiology image, predict relevant biomedical terms (tags).

Predict tags…

Ultrasonography, Ectopic kidney, Pelvis, Kidney, Calculi, Obstructed

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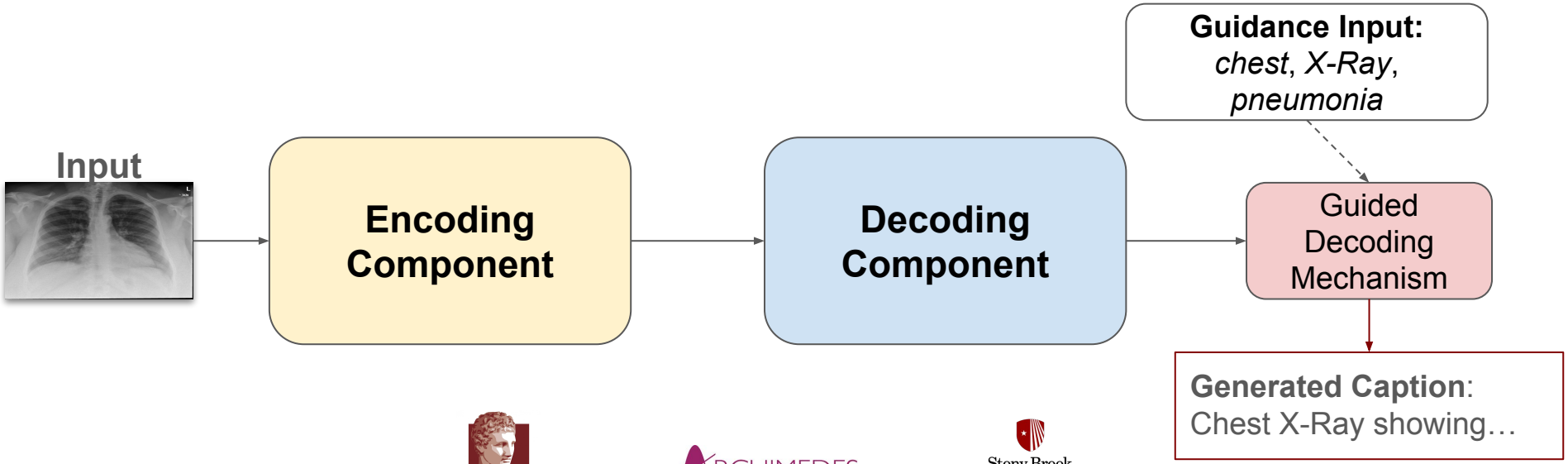
DMMCS: A tag-based guided decoding mechanism

- In text generation, decoding is the process of converting the model's probabilistic output into coherent and meaningful text.
- **Guided Decoding**: A technique to steer the text generation of language models towards desired outcomes or characteristics.



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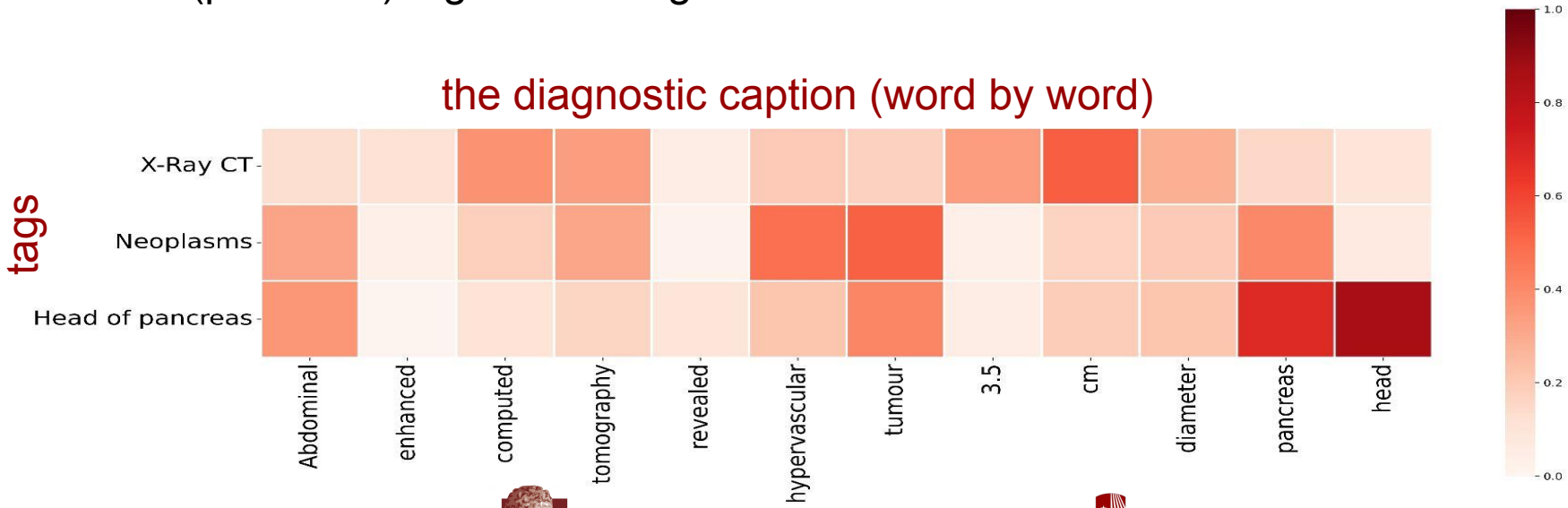
DMMCS: Exploratory Analysis

Idea: The quality of a diagnostic caption is strongly associated with how well the generated report represents the key biomedical tags associated with the medical image.

Exploration Step: Measure the cosine similarity between each word of the caption and each (predicted) tag of the image.



DMMCS: Definitions

Next step: investigate the correlation between a tag and all the captions it is associated with in the training set.

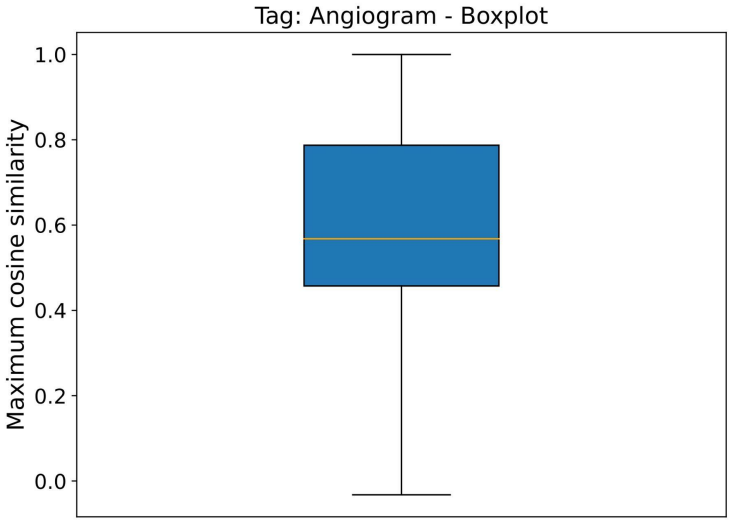
- For a caption s and a tag t, the maximum cosine similarity (**MCS)** shows how strongly t is represented in the tokens of s
- The median of MCS (**MMCS**) shows how strongly t is represented (on average) across captions it is associated with

DMMCS: Maximum Cosine Similarity (MCS)

Question: Are all the biomedical tags always strongly mentioned in the captions with which they are associated?

how strongly tag *"Angiogram"* is mentioned (on average) on training captions it is associated with.

Answer: No!

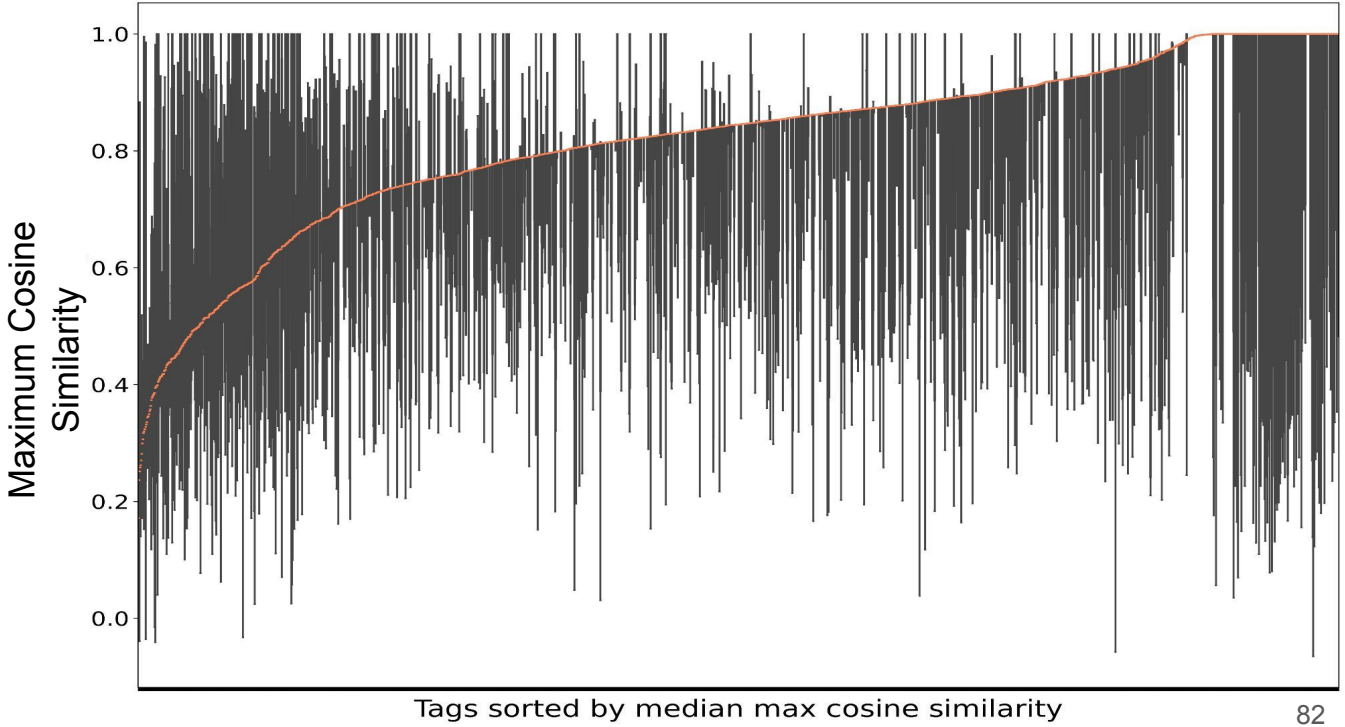


DMMCS: Median Maximum Cosine Similarity (MMCS)

(Another) question: Do all of the tags follow an MCS distribution similar to the tag *"Angiogram"*?

- The orange line denotes the median value of the MCS distribution per tag

Observation: The MMCS values can vary a lot between different tags.



DMMCS: Distance from Median Maximum Cosine Similarity

Goal: At each decoding step, incentivize the decoder to choose captions that encapsulate the associated biomedical tags more or less explicitly in the diagnostic caption.

➔ The level of explicitness depends on the MMCS values of each tag

How?

➔ Impose a penalty at each decoding step, given tags T and caption s:

$$DMMCS_p(T, s) = \frac{1}{|T|} \cdot \sum_{t \in T} (MCS(t, s) - MMCS(t))^2$$

We want the sum to be as close to zero as possible!

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DMMCS: Scoring Formula

Overall, at each decoding step, the beam search algorithm keeps the n best candidate captions according to the following score:

$$DMMCS(s) = \alpha \cdot DMMCS_p(T, s) + (1 - \alpha) \cdot (1 - D_{score})$$

← *the logprob*

<u>Formula breakdown:</u>

- $\boldsymbol{\alpha}$: A fixed weighting factor that regulates the effect of the two addends.
- $\boldsymbol{D_{score}}$: Denotes the typical LM score used in Beam Search.

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DMMCS: Experimental Scenarios

- We evaluated the proposed method on **four captioning models** and **two biomedical datasets**.

- Models:
  - Show and Tell (CNN-RNN)
  - ViT-GPT2
  - InstructBLIP
  - OpenFlamingo

- Datasets
  - ImageCLEFmedical 2023
  - MIMIC-CXR

- Evaluation Metrics
  - BLEU
  - BLEURT
  - Clinical Accuracy (CA)
  - Perplexity

- Baselines
  - Vanilla Beam Search
  - Strict Constrained Beam Search
  - Disjunctive Constrained Beam Search

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DMMCS: Evaluation (3)

ImageCLEFmedical 2023 Dataset - With Predicted Tags								
	BLEU				BLEURT			
	BS	*ConBS* ∀	*ConBS* ∃	*DMMCS*	*BS*	*ConBS* ∀	*ConBS* ∃	*DMMCS*
Show & Tell	20.61 (0.33)	20.52 (0.39)	21.21 (0.38)	**21.27** (0.35)	29.99 (0.14)	30.03 (0.17)	30.39 (0.13)	**30.47** (0.08)
ViT-GPT2	15.34 (0.09)	15.75 (0.12)	16.29 (0.08)	**16.31** (0.08)	26.50 (0.09)	26.31 (0.10)	26.92 (0.14)	**27.01** (0.16)
InstructBLIP	11.81 (0.09)	15.89 (0.08)	**16.14** (0.13)	15.93 (0.11)	29.68 (0.26)	29.71 (0.12)	30.08 (0.14)	**30.10** (0.15)
Flamingo	15.34 (0.11)	15.81 (0.13)	**15.92** (0.06)	15.47 (0.09)	28.49 (0.19)	30.11 (0.21)	30.67 (0.19)	**31.34** (0.16)

MIMIC-CXR Dataset - With Predicted Tags								
	BLEU				BLEURT			
	BS	*ConBS* ∀	*ConBS* ∃	*DMMCS*	*BS*	*ConBS* ∀	*ConBS* ∃	*DMMCS*
Show & Tell	11.77 (0.18)	12.14 (0.22)	13.38 (0.17)	**14.13** (0.24)	29.08 (0.27)	29.04 (0.29)	29.21 (0.31)	**29.49** (0.29)
ViT-GPT2	13.55 (0.26)	12.91 (0.19)	13.67 (0.23)	**14.78** (0.21)	23.52 (0.34)	24.87 (0.37)	**24.98** (0.30)	24.37 (0.29)
InstructBLIP	12.76 (0.20)	12.09 (0.22)	12.27 (0.17)	**13.32** (0.19)	24.65 (0.24)	26.28 (0.27)	**26.43** (0.26)	25.56 (0.25)
Flamingo	12.78 (0.11)	13.07 (0.13)	**13.39** (0.11)	13.26 (0.16)	29.14 (0.22)	28.86 (0.21)	29.58 (0.27)	**29.81** (0.24)

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DMMCS: Qualitative Results

Gold Caption	Vanilla Beam Search Decoding	DMMCS Decoding	Predicted Tags
Echocardiogram showing trivial **pericardial effusion.**	**Parasternal long axis view** of the **left ventricular outflow tract**. The arrow indicates the position of the left ventricular outflow tract.	Transthoracic **echocardiography** parasternal long axis view showing **pericardial effusion**.	*Transthoracic Echocardiography* *Pericardial Effusion*
Ultrasound of the **testes** showing a left testicular **mass** (encompassed within the yellow plus symbols).	**Ultrasound** scan of the right **breast** showing a well defined **lesion** with irregular margins.	**Ultrasonography** of the *right* **testis** showing a heterogeneous hypoechoic **mass** in the lower pole of the right **testis**.	*Ultrasonography* *Ultrasonography - Doppler - Duplex* *Structure of right testis*

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DMMCS: Performance analysis

- Computational Overhead:
  - 17-19% **temporal** overhead (Update!)
  - 4-6% **memory usage** overhead

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